Networking Fundamentals 4CC523

Week 7 IP Addressing

Individual Health & Safety Responsibilities

To help prevent the spread of the COVID-19 virus we ask you to carry out the following, to help keep everyone safe & aid with contact tracing.

- **If you have any COVID-19 symptoms leave the class immediately if possible & return to your accommodation. Contact the NHS on 111 & then contact student wellbeing.**
- **Wash your hands or sanitise them upon entering/exiting the area.**
- **Sign in/register via the Tap In system or alternative system in place.**
- **Identify a seating area & log the desk number on 'Who Sat Where?' on the Udo app. For people who don't have access to the app inform your lecturer to use the alternative system.**
- **Always maintain social distancing.**
- **When the class finishes, please use the cleaning materials provided to clean down the area you have used & then wash or sanitise your hands.**
- **We welcome the wearing of face coverings in teaching areas where possible.**

Thanks for your help and continued support.

IMPORTANT INFO!!

- BE AWARE - YOU ARE BEING RECORDED

  This is part of the normal lecture process, recordings will be made available to you to support your studies

  Inform me as soon as possible if you object to recordings being made (and remind me every session!!)

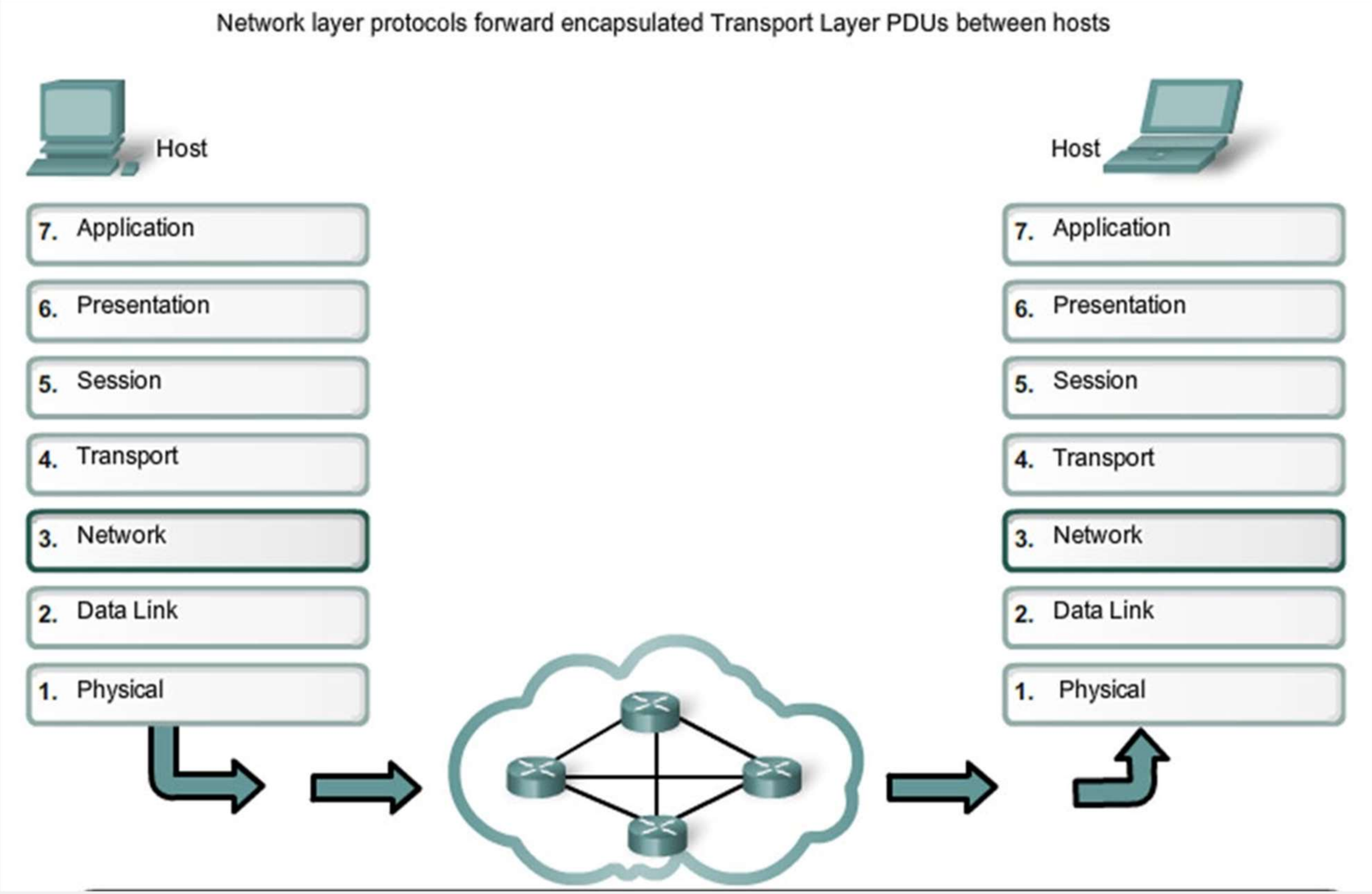
- DID YOU TAP-IN?? DO IT NOW!

Week 7: Objectives

- Network Layer Protocol - IP
- IPv4 Network Addresses
- IPv6 Network Addresses
- Connectivity Verification
- Summary

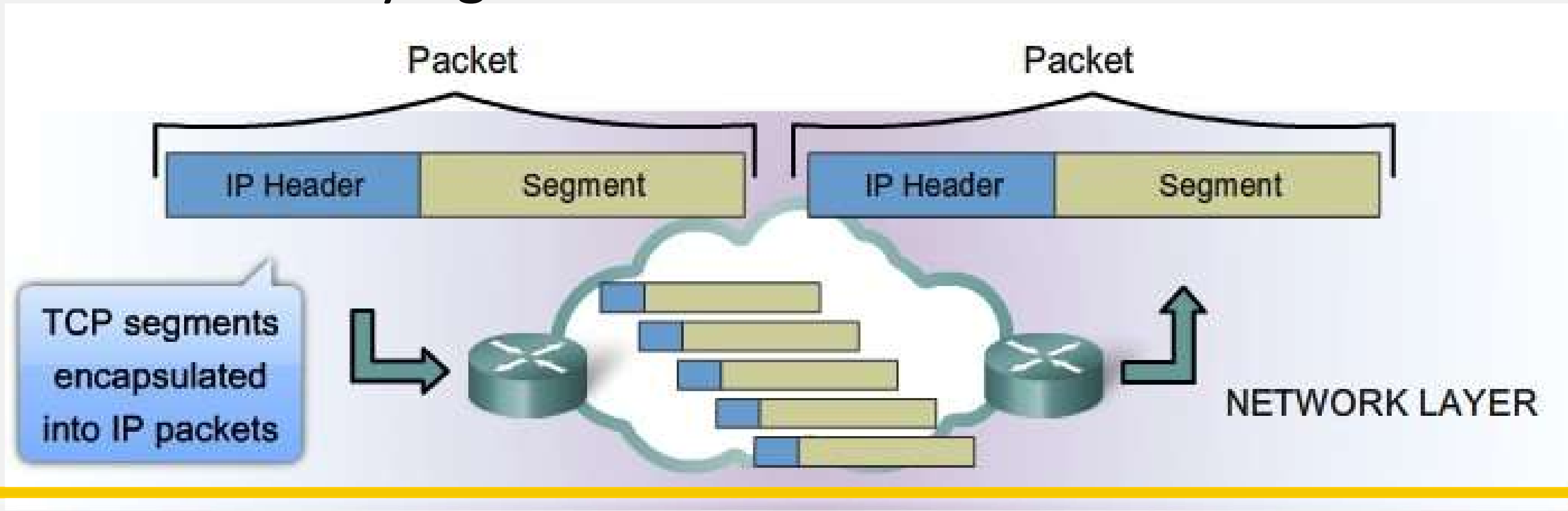
- This is going to be a fast one!!!

Network Layer in Communication



Characteristics of IP

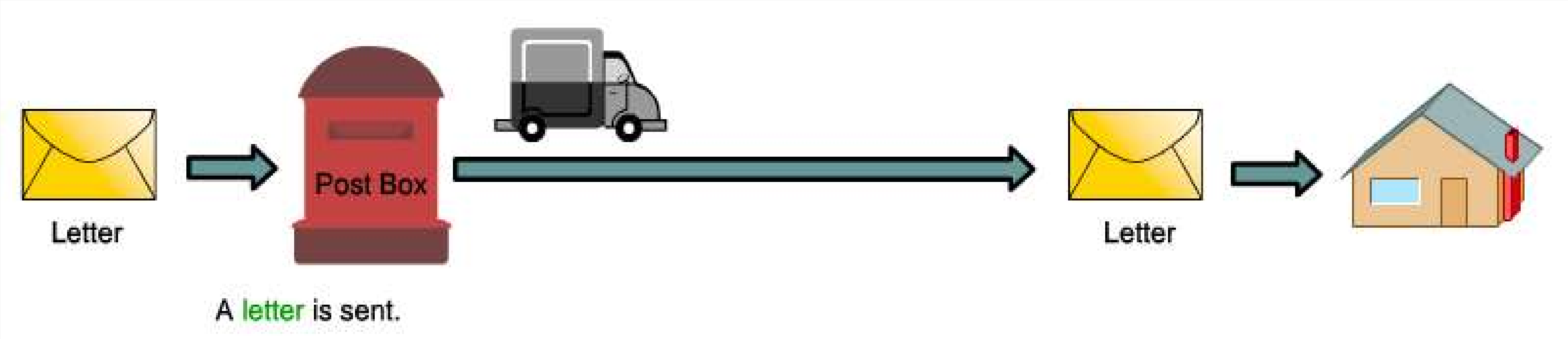
- “Connectionless” – No connection is established before sending data packets
- Best Effort (unreliable) delivery – No overhead is used to guarantee packet delivery
- Media independent – operates independently of the medium carrying the data



IP - Connectionless

- This is nothing to do with Wired vs Wireless
- This has nothing to do with physical connection
- Physical connection was at layer 1, we are looking at layer 3
- Layer 3 is where *Logical* addressing happens
- This **<u>is</u>** about (not) checking if the *intended recipient* of your message *is listening* (or even exists) *before* you start talking

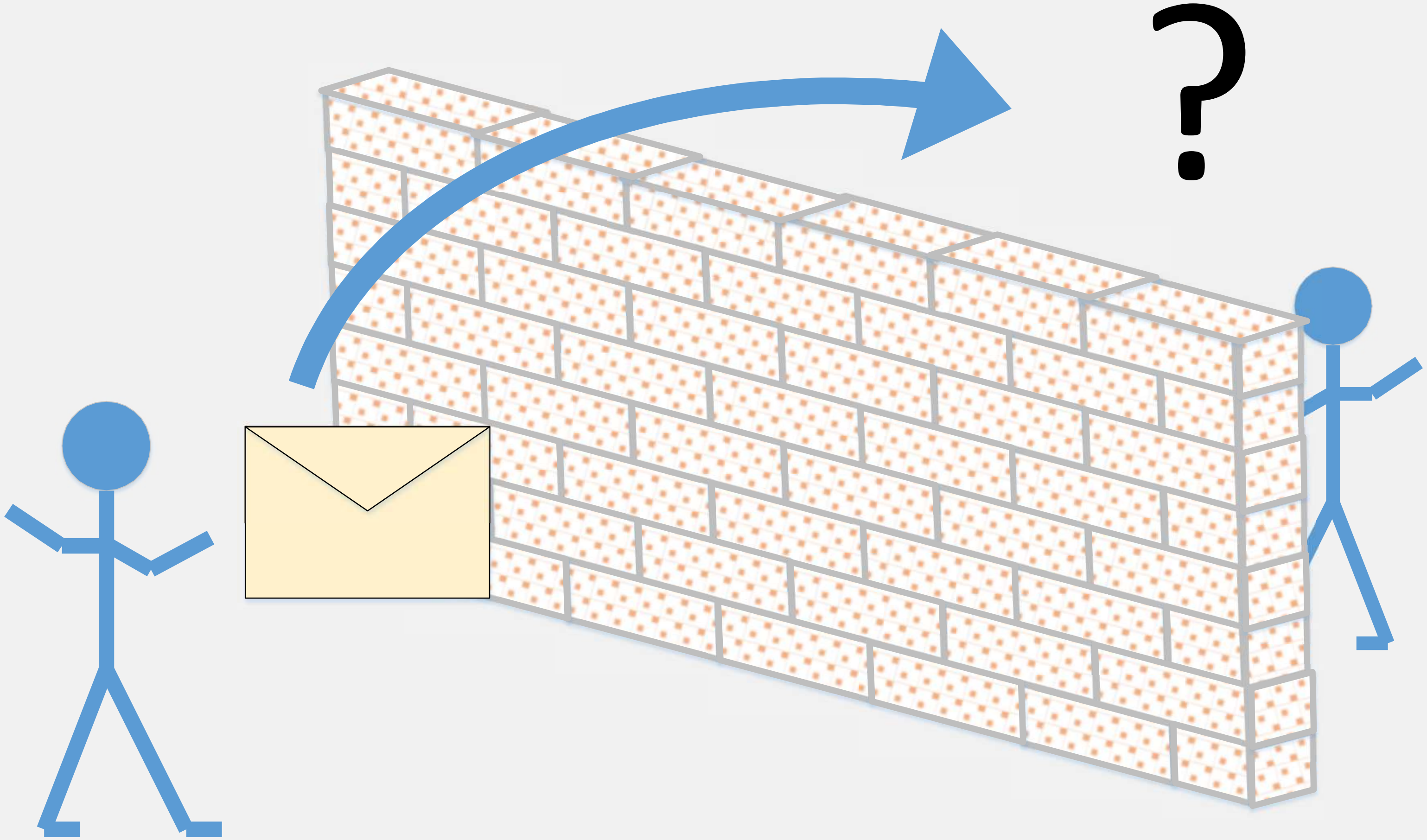
IP – Connectionless example



- Sender knows:
  - the letter went
- Sender does not know if it:
  - arrived
  - was thrown in the bin
  - was opened
  - was read
  - was replied to
- Intended receiver doesn't know:
  - When it might arrive
  - If a letter is arriving
  - If the sender sent anything

And then there could be a return path too with the same issues…

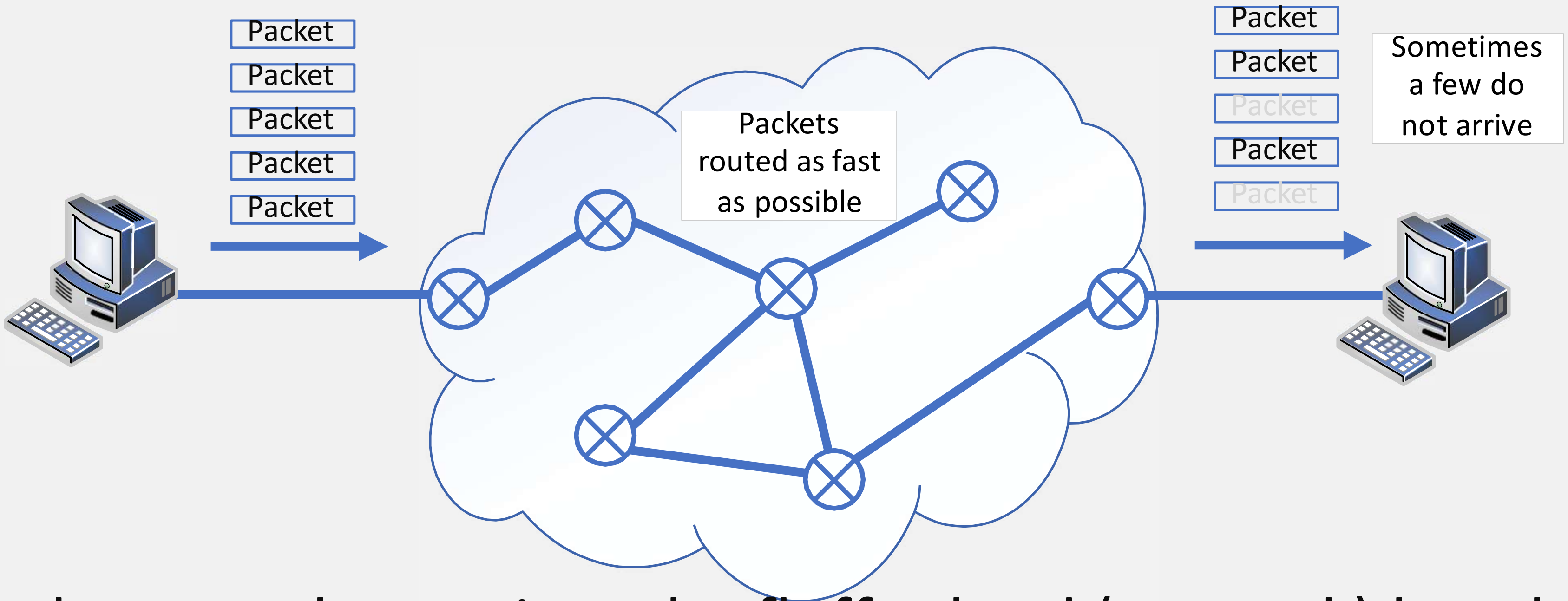
IP – Connectionless
(an alternative example)



IP – Best Effort Delivery

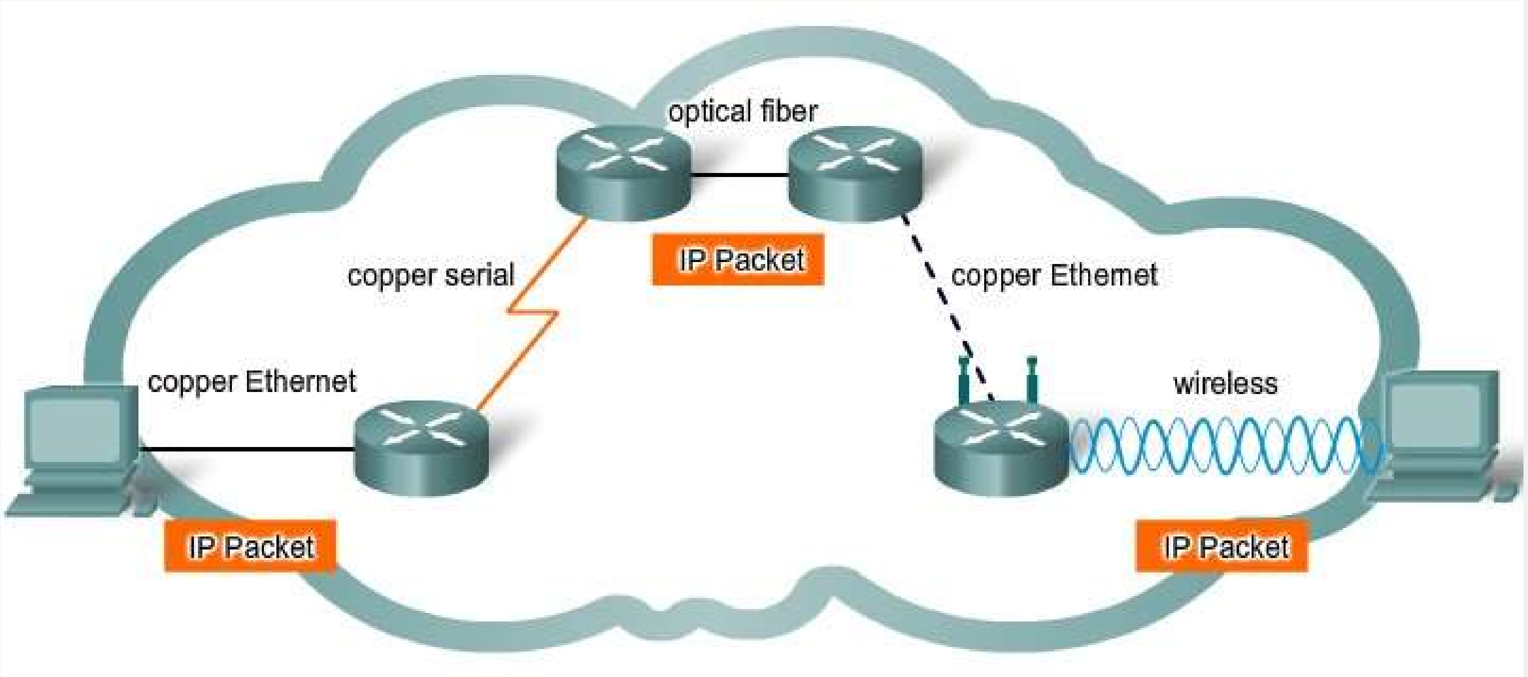
- Packets are sent and the carrier does their best but accidents happen
- Some might be dropped or lost (or sent to the wrong place)
- IP is described as “Unreliable” transport

IP – Best Effort Delivery



- Packets are thrown into the fluffy cloud (network) but the sender has no idea if they arrive at the destination or not
- This IP thing is starting to sound a bit rubbish, how does it ever work???

IP – Media Independent



- The layers model means that IP (at layer 3) does not need to consider (care about) the layer 1 physical medium, any practical medium can be used
- At last there is something positive to say about IP!

How does it ever work?

- Not giving too much away but

- Sometimes it does not matter that data is lost
- Sometimes it matters but it was time sensitive and the useful time has passed

- The layers above 3 might be able to help…

Encapsulation (again)

- As a general rule, when data is passed from upper to lower layers headers get added

- IP layer is no different, a **segment** from layer 4 becomes the data for a **packet** at layer 3

De-encapsulation (again)

- When an IP layer 3 packet arrives at a destination, the header is stripped off and the data then becomes a transport layer (layer 4) segment again with its own header

IPv4 Packet Header – 20 Bytes (normally)

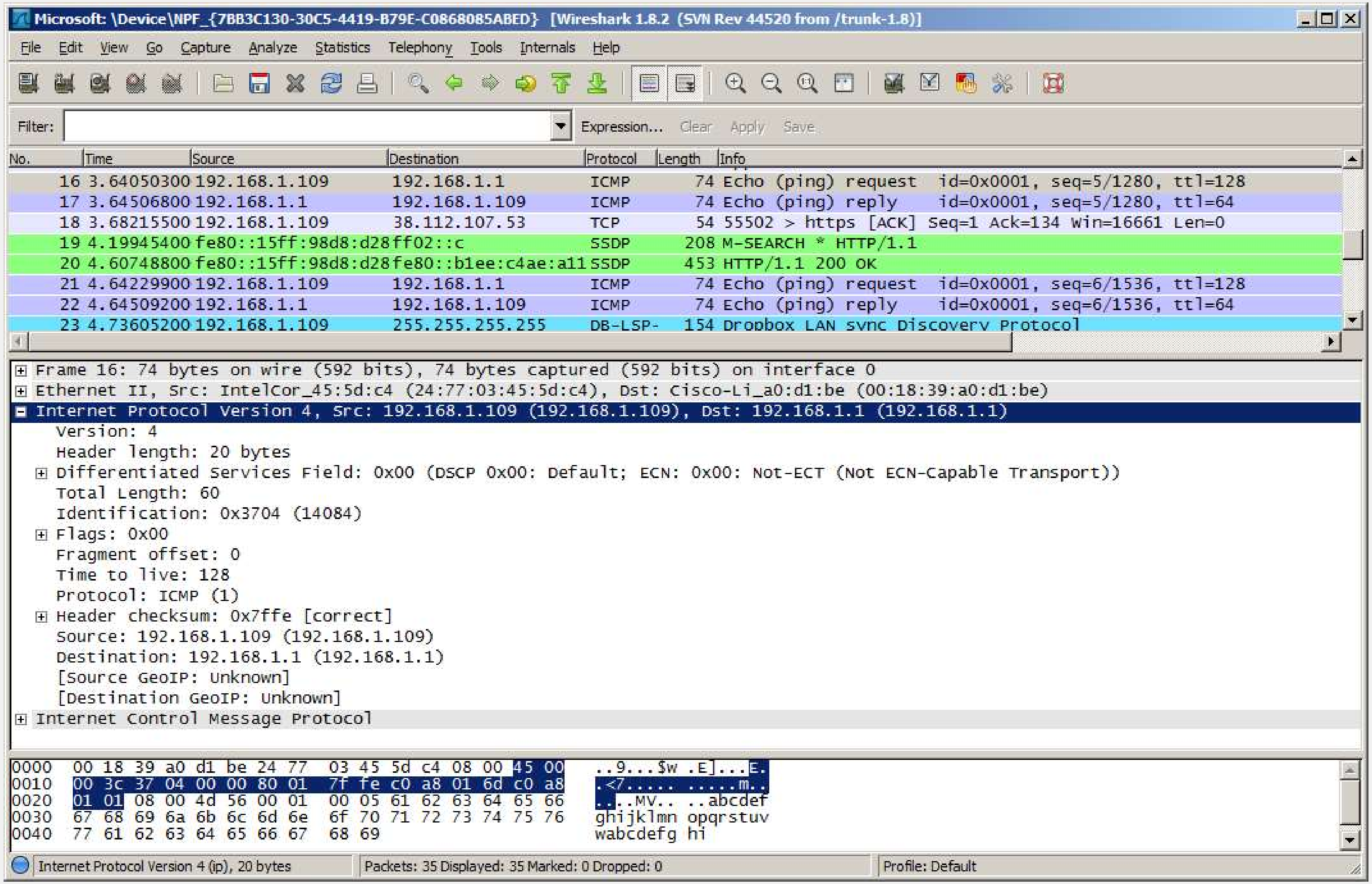
<table>
<tr><th colspan="2">Byte 1</th><th colspan="2">Byte 2</th><th colspan="2">Byte 3</th><th colspan="2">Byte 4</th></tr>
<tr><td rowspan="2">Version</td><td rowspan="2">IP Header Length</td><td colspan="2">Differentiated Services</td><td colspan="4" rowspan="2">Total Length</td></tr>
<tr><td>DSCP</td><td>ECN</td></tr>
<tr><td colspan="4">Identification</td><td>Flag</td><td colspan="3">Fragment Offset</td></tr>
<tr><td colspan="2">Time To Live</td><td colspan="2">Protocol</td><td colspan="4">Header Checksum</td></tr>
<tr><td colspan="8">Source IP Address</td></tr>
<tr><td colspan="8">Destination IP Address</td></tr>
<tr><td colspan="7">Options (optional - IHL >5 if options are present)</td><td>Padding</td></tr>
</table>

Note that there is no hardware addressing – that header has not been added yet

IPv4 Header – jfi (in the notes):

- Version – four bits, in IPv4, this is always equal to 4.
- Internet Header Length (IHL) - 4 bits indicating number of 32-bit words. Min value = 5 (no options, header=20 bytes) Max = 15 (10 byte pairs of options, header =60 bytes)
- Padding – fills the remaining space if the options use less than a byte pair (word)
- Differentiated Services Code Point (DSCP) - Originally "type of service" (ToS), now specifies differentiated services (DiffServ) real-time data streaming technologies (e.g. Voice over IP).
- Explicit Congestion Notification (ECN) – Optional; Allows end-to-end network congestion notification if both endpoints are capable and agree to use it; effective only when supported by the underlying network.
- Total Length – 16 bits defines the entire ip packet size in bytes (including header and data). Min size = 20 bytes (header without data), max = 65,535 bytes. Hosts must be able to reassemble datagrams of up to 576 bytes, many handle much larger packets. Datagrams are fragmented at 1500 bytes
- Identification – 16 bits normally used to uniquely identify fragments of a single IP datagram.
- Flags – 3 bits used to control or identify fragments. bit 0: Reserved; must be zero, bit 1: Don't Fragment (DF), bit 2: More Fragments (MF). If DF is set but fragmentation is required the packet gets dropped. Can be used for hosts that do not have resources to handle fragmentation and for path MTU discovery. MF is set for fragmented packets, except the last fragment. The last fragment has a non-zero Fragment Offset field, differentiating it from an unfragmented packet.
- Fragment Offset – 13 bits measuring units of eight-byte blocks. Specifies the offset of a fragment relative to the beginning of the original unfragmented IP datagram, first fragment has an offset of zero. Max offset of (213 – 1) × 8 = 65,528 bytes – exceeds max IP packet length of 65,535 bytes when header is added (65,528 + 20 = 65,548 bytes).
- Time To Live (TTL) – 8 bit number limiting the hopcount of a datagram preventing a packet from persisting forever (e.g. going in circles) on an internet. TTL reduces by 1 at each when ttl = zero, packet is dropped (normally returns an ICMP Time Exceeded message)
- Protocol – 8 bits, identifies the protocol of the data portion of the IP datagram. IANA allocates the list of IP protocol numbers
- Header Checksum – 16 bits used for error-checking of the header, When a packet arrives at a router a new checksum is calculated because the TTL decreases

Sample IPv4 Header - Wireshark



Sample IPv4 Header - Wireshark

```
⊞ Frame 16: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
⊞ Ethernet II, Src: IntelCor_45:5d:c4 (24:77:03:45:5d:c4), Dst: Cisco-Li_a0:d1:be (00:18:39:a0:d1:be)
⊟ Internet Protocol Version 4, Src: 192.168.1.109 (192.168.1.109), Dst: 192.168.1.1 (192.168.1.1)
    Version: 4
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
    Total Length: 60
    Identification: 0x3704 (14084)
  ⊞ Flags: 0x00
    Fragment offset: 0
    Time to live: 128
    Protocol: ICMP (1)
  ⊞ Header checksum: 0x7ffe [correct]
    Source: 192.168.1.109 (192.168.1.109)
    Destination: 192.168.1.1 (192.168.1.1)
    [Source GeoIP: Unknown]
    [Destination GeoIP: Unknown]
⊞ Internet Control Message Protocol

0000  00 18 39 a0 d1 be 24 77  03 45 5d c4 08 00 45 00   ..9...$w .E]...E.
0010  00 3c 37 04 00 00 80 01  7f fe c0 a8 01 6d c0 a8   .<7..... .....m..
0020  01 01 08 00 4d 56 00 01  00 05 61 62 63 64 65 66   ....MV.. ..abcdef
0030  67 68 69 6a 6b 6c 6d 6e  6f 70 71 72 73 74 75 76   ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67  68 69                     wabcdefg hi

Internet Protocol Version 4 (ip), 20 bytes | Packets: 35 Displayed: 35 Marked: 0 Dropped: 0 | Profile: Default
```

IPv4 Addresses

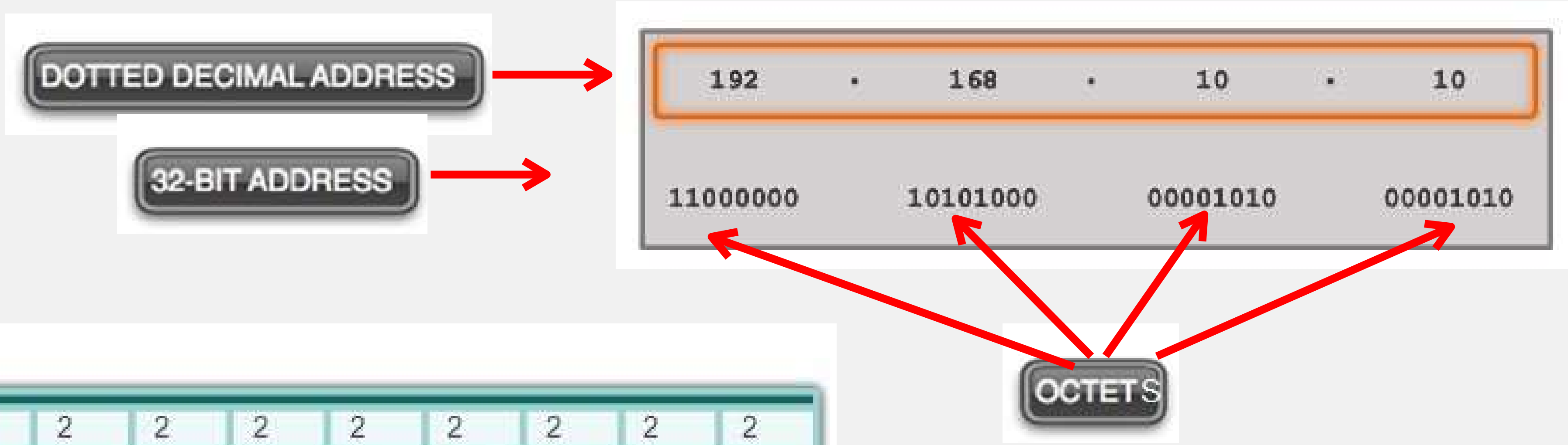
- IPv4 addresses are in 4 sections of Dotted Decimals
- Each section is an **octet** or 8 bit binary string which limits the range of numbers to 256 values between 0 and 255 inclusive
- IPv4 addresses identify a network (LAN, Subnet, Cloud) and a host (individual interface on a device)

- Octets are decimal representations of 8 bit binary strings (bytes)

IPv4 Addresses

- The **network address** is the same for all devices that are part of the same network
- The **host address** identifies a specific device (or interface) within the network
- A **subnet mask** is used to identify the network address from a given IP address and identify the size of the network (max number of hosts)
- The **subnet mask** will be the same for all devices that are part of the same network
- Devices on the same network can communicate directly, devices in different networks need a Layer 3 device, such as a router

Binary Number System



Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Octet Bit Values	128	64	32	16	8	4	2	1
Binary Address	1	1	0	0	0	0	0	0
Binary Bit Values	128	64	0	0	0	0	0	0

Add the binary bit values. 128 + 64 = 192

Note: 8 bits to an octet,
8 bits to each "decimal" portion
Smallest octet = "all zeros"
= decimal 0
Largest octet = "all ones"
= decimal 255

Converting a Binary Address to Decimal

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	0	1	1	0	0	0	0

128 + 32 = 160, +16 = 176

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1

128 + 64 + 32 + 16 + 8 +4 + 2 + 1 = 255

OR 256 -1 = 255

OR "all ones" = 255

Converting from Decimal to Binary

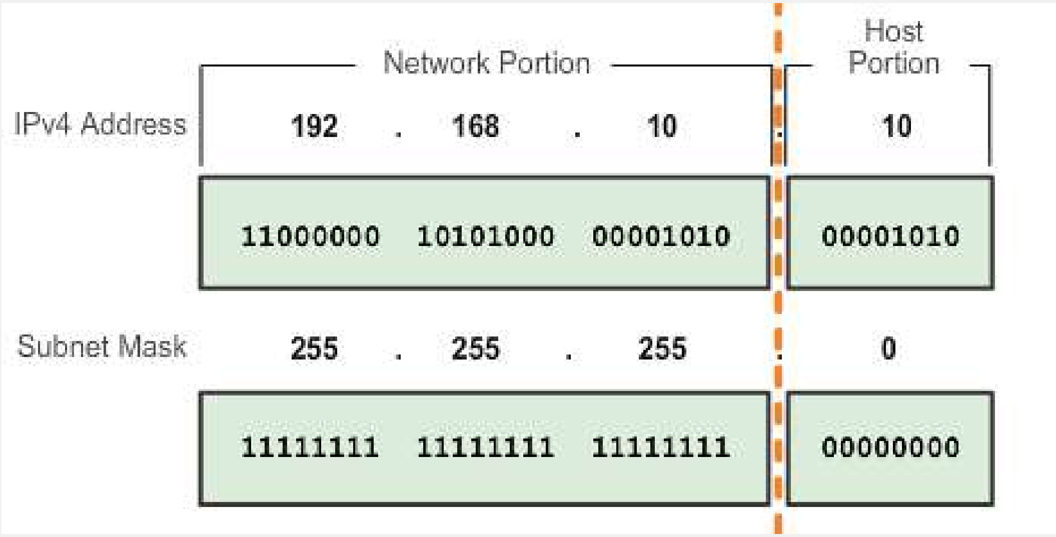
Convert Decimal to Binary

192.168.10.10

11000000 **10101000**

		128	64	32	16	8	4	2	1
168 -128	> 128, place a 1 in the 128 position subtract 128	1							
40	< 64, place a 0 in the 64 position do not subtract	1	0						
40 -32	> 32, place a 1 in the 32 position subtract 32	1	0	1					
8	< 16, place a 0 in the 16 position do not subtract	1	0	1	0				
8	= 8, place a 1 in the 8 position subtract 8	1	0	1	0	1			
0	place a 0 in all remaining positions All done. Result	**1**	**0**	**1**	**0**	**1**	**0**	**0**	**0**

IPv4 Address structure



- Logical AND of the IP address and the subnet mask will show the network address
- The subnet mask often contains one or more 255 sections
- It is sometimes identified as a "slash number" which refers to the number of ones e.g. "a 24 bit mask" or /24 = 255.255.255.0

Binary AND operator

- AND : Answer is 1 (true) only when ALL inputs are 1 (true)

  1 AND 1 = 1, 1 AND 0 = 0, 0 AND 1 = 0, 0 AND 0 = 0

  (think of this as multiplication)

- For info:
- OR : true (1) when ANY input is true (1)

  1 OR 1 = 1, 1 OR 0 = 1, 0 OR 1 = 1, 0 OR 0 = 0

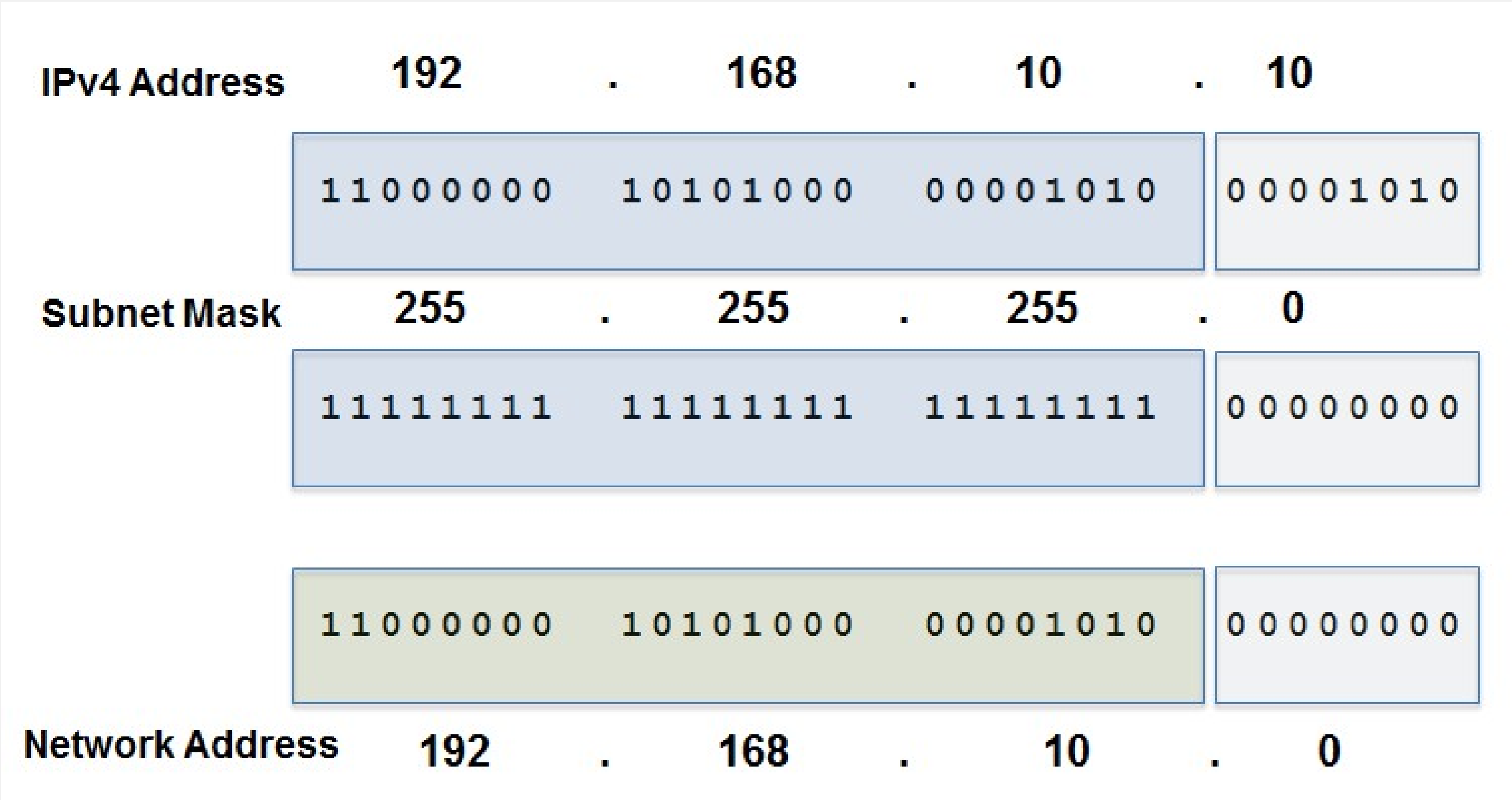
  0 (false) only if everything is 0 (false)

- XOR (exclusive or) : true (1) when ONLY ONE input is true (1)

  1 XOR 1 = 0, 1 XOR 0 = 1, 0 XOR 1 = 1, 0 XOR 0 = 0

  Finds individuals in a crowd!

Bitwise AND Operation



1 AND 1 = 1   1 AND 0 = 0   0 AND 1 = 0   0 AND 0 = 0

Legacy Classful Addressing

- Originally we determined the size of the network (LAN, subnet, cloud) according to the binary value of the first octet
- 0xxx xxxxb is "Class A" and has a subnet mask of 255.0.0.0 (aka "slash 8" or "/8" or "8 bit mask")
- 10xx xxxxb is "Class B" and has a subnet mask of 255.255.0.0 (aka "slash 16" or "/16" or "16 bit mask")
- 110x xxxxb is "Class C" and has a subnet mask of 255.255.255.0 (aka "slash 24" or "/24" or "24bit mask")

Legacy Classful Addressing

Address Class	1st octet range (decimal)	1st octet bits (green bits do not change)	Network(N) and Host(H) parts of address	Default subnet mask (decimal and binary)	Number of possible networks and hosts per network
A	1-127**	00000000-01111111	N.H.H.H	255.0.0.0	128 nets (2^7) 16,777,214 hosts per net (2^24-2)
B	128-191	10000000-10111111	N.N.H.H	255.255.0.0	16,384 nets (2^14) 65,534 hosts per net (2^16-2)
C	192-223	11000000-11011111	N.N.N.H	255.255.255.0	2,097,150 nets (2^21) 254 hosts per net (2^8-2)
D	224-239	11100000-11101111	NA (multicast)		
E	240-255	11110000-11111111	NA (experimental)		

Legacy Classful Addressing

- Class A begin [0 – 127].x.x.x and have 2^24 addresses per subnet (2 ^ number of zero bits in the mask) 16,777,216
- Class B begin [128 – 191].x.x.x and have 2^16 addresses per subnet (2 ^ number of zero bits in the mask) 65536
- Class C begin [192 – 223].x.x.x and have 2^8 addresses per subnet (2 ^ number of zero bits in the mask) 256

Host Addressing

- Values stated disagree with the table (class C in table = 254 addresses, calculated 256) – why?

- The first address in any network defines the network, this is the host address where the binary equivalent is "all zeros"
  - e.g. network.0.0.0 /8
  - e.g. network.network.network.0 /24
- The network definition cannot be used as a host address – it is not a useable address

Host Addressing

- Values stated disagree with the table (class C in table = 254 addresses, calculated 256) – why?

- The last address in any network is a broadcast to "everything in the network", this is the host address where the binary equivalent is "all ones"
  - e.g. network.255.255.255 /8
  - e.g. network.network.network.255 /24
- The Broadcast Address cannot be used as a host address – it is not a useable address

Network, Host, Broadcast Addresses

Given an IP address of 10.1.1.10 and a mask of 255.255.255.0, we can find the network, host and broadcast addresses by calculating Network Address AND Subnet Mask then reading the host number or setting the host to zero or one.

Network Portion			Host Portion	
10	1	1	0	
00001010	00000001	00000001	00000000	All 0s – NETWORK ADDRESS
10	1	1	10	
00001010	00000001	00000001	00001010	0s and 1s in host portion
10	1	1	255	
00001010	00000001	00000001	11111111	All 1s – BROADCAST ADDRESS

Network, Host and Broadcast
(some hints for you to read)

- The Network and Broadcast addresses cannot be used by individual hosts
- Using “all ones” means that the “last bit” of the last octet must be a one and as a result the equivalent decimal number is “odd” (always)
- The Broadcast is the “all ones address” and must be an odd number
  - not all odd numbers are broadcast addresses but broadcast addresses are always odd numbers
  - The **last usable host address** is the broadcast address – 1, so the last host address must always be an even number

- “All zeros” means that the last bit is always zero meaning the decimal equivalent must be “even” (always)
- The network address is the “all zeros address”
  - an odd number cannot be a network address but not all even numbered addresses are network addresses
  - The **first usable host address** is the network address + 1, so the first host must be odd.

Network, Host and Broadcast
(some hints for you to read)

- The binary representation of a subnet mask always has ones to the left, zeros to the right. The only odd number that can be part of a subnet mask is the “all ones” for an octet (255)
- If a subnet mask octet contains 255, the next octet could be a range of numbers, if an octet contains any number other than 255, all subsequent octets MUST be zero
  - Any number other than 255 must have a last bit of zero, all subsequent bits must be zero.
- The address zero exists.
  - A 2 bit can be 00b to 11b (0d to 3d) - four addresses not three!

IPv4 transmission types

- Unicast – between two individual devices
- Broadcast – from one device to many
- Multicast – from one to a group of some devices

“Public” IPv4 Addresses

- Hosts that require direct access to the Internet must be uniquely identified
- Unique identity is known as a “public IP address”
- Lists of who has each IP address are held by IANA – the Internet Assigned Numbers Authority
- Management is split between 5 Regional Internet Registries (RIRs)
  - APNIC, ARIN, RIPE NCC, LACNIC and AFRINIC.

“Private” IPv4 Addresses

- Private addresses cannot be used in the Internet and are not unique, they are reused continually
- The most common class A, B and C private ranges are:

- 10.0.0.0 to 10.255.255.255 (10.0.0.0/8)
- 172.16.0.0 to 172.31.255.255 (172.16.0.0**/12**)
- 192.168.0.0 to 192.168.255.255 (192.168.0.0**/16**)

- These are definitions, it is what they are, you cannot change them!

Other “Private” IPv4 Addresses

- 127.0.0.0/8 (this refers to “me”. 127.0.0.1 should always answer but you are talking to yourself)

- Auto-IP or Automatic Private IP Addressing (APIPA) uses 169.254.1.0 – 169.254.254.255 (aka "local link addressing")
  - Technically 169.254.0.0 /16 but not 169.254.0.0 – .255 or 169.254.255.0 –.255
  - Allocated to a device by itself when a DHCP server does not answer

NAT and PAT

- Privately addressed devices can access the Internet through NAT and/or PAT
- Network Address Translation can be used by a firewall or other edge device to convert a “private” to a “public” IP address
- Port Address Translation (or NAT overload) can be used to convert **many private IP** addresses **to a single public address**

IPv4 problems

- There are around 4 billion IPv4 addresses available (4,000,000,000)
- There are around 7.5 billion people on and around Earth, many want more than one IP address
- IP Address depletion refers to there not being enough IP addresses for everyone to use
- Some people think that NAT causes a problem because it prevents direct end-to-end connectivity

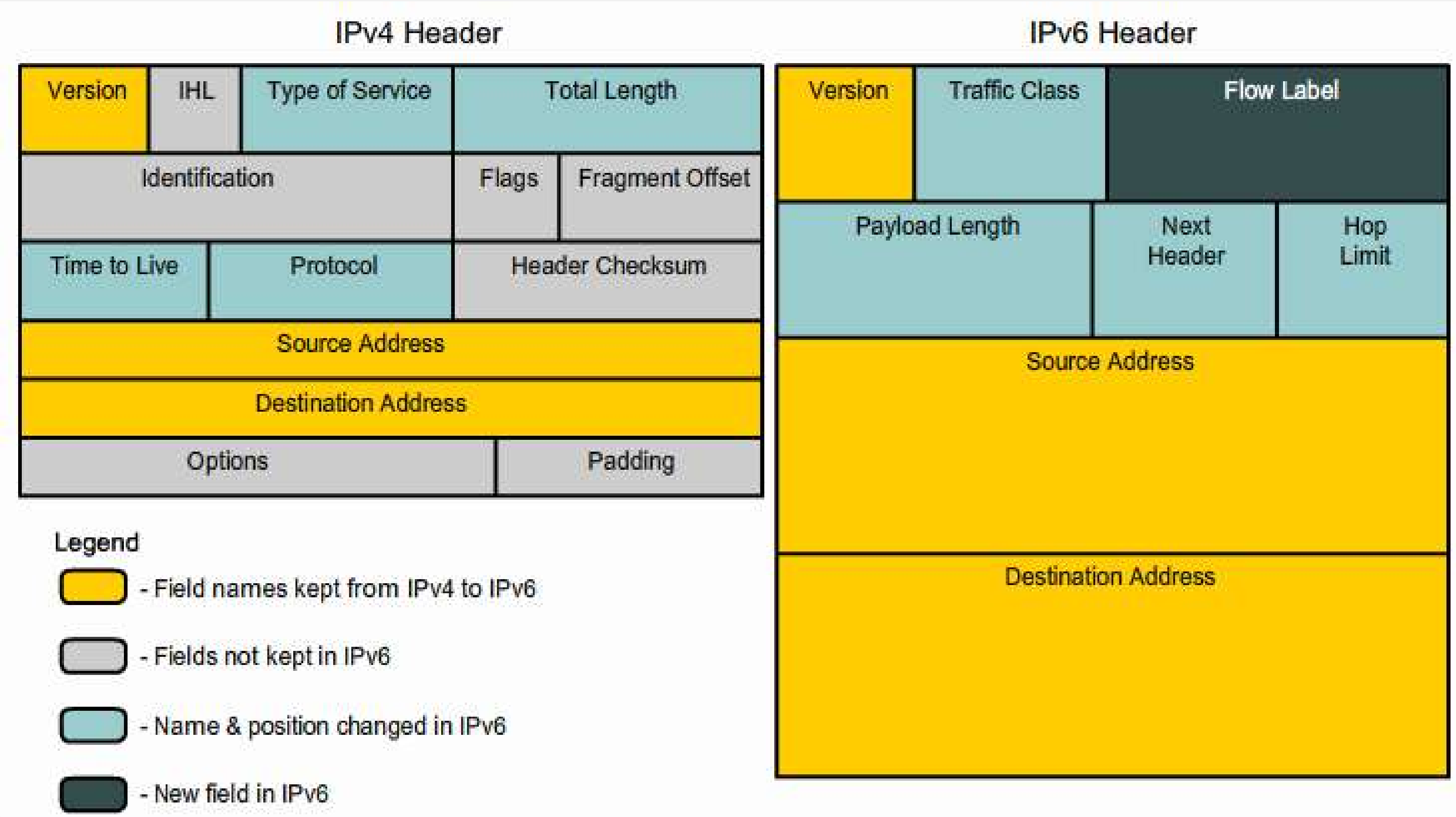
Introducing IPv6

- 340 undecillion addresses
340,000,000,000,000,000,000,000,000,000,000,000,000
  - (36 zeros, 12 groups of 3)

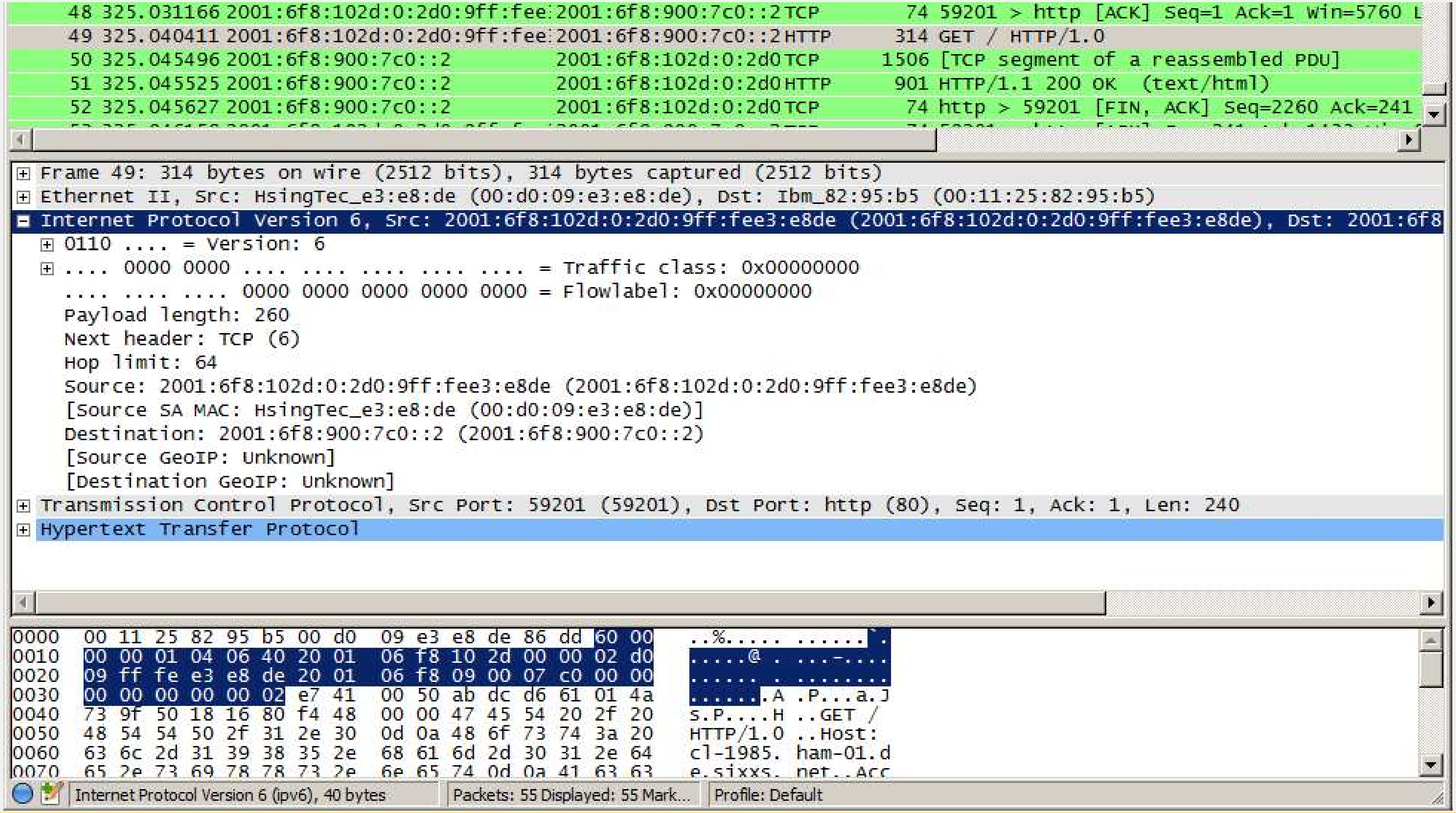
- The intention of IPv6 is to replace IPv4
  - Increased address space
  - Improved packet handling
  - Eliminates the need for NAT
  - Integrated security

IPv6 Packet Header

- Always 40bytes, addresses are 16 bytes (each)



Sample IPv6 Header



IPv6 Address Representation

- Hex numbering system uses the numbers 0 to 9 and the letters A to F
- Consistent 4 digit groups
- Four bits are represented with a single hexadecimal value

Hexadecimal	Decimal	Binary
00	0	0000 0000
01	1	0000 0001
02	2	0000 0010
03	3	0000 0011
04	4	0000 0100
05	5	0000 0101
06	6	0000 0110
07	7	0000 0111
08	8	0000 1000
0A	10	0000 1010
0F	15	0000 1111
10	16	0001 0000
20	32	0010 0000
40	64	0100 0000
80	128	1000 0000
C0	192	1100 0000
CA	202	1100 1010
F0	240	1111 0000
FF	255	1111 1111

IPv6 Address Representation

- 128 bits in length and written as a string of hexadecimal values
  - 2001:0DB8:0000:1111:0000:0000:0000:0200
  - FE80:0000:0000:0000:0123:4567:89AB:CDEF
  - (4 bits to each character, 2 characters per byte, 2 bytes per section)
- Can be written in either lowercase or uppercase
- 2001:0DB8::/32 is a “documentation address”, not real, used for examples, don’t copy it in real life (use it in labs) (could be “start 2001, zero debate!”)
- Real public numbers start 2::/3 (001x xxxx xxxx xxxx:)

:: - The IPv6 compressed format

Preferred	2001:0DB8:000A:1000:0000:0000:0000:0100
No leading 0s	2001: DB8: A:1000: 0: 0: 0: 100
Compressed	2001:DB8:A:1000:0:0:0:100

Preferred	2001:0DB8:0000:0000:ABCD:0000:0000:0100
Omit leading 0s	2001: DB8: 0: 0:ABCD: 0: 0: 100
Compressed	2001:DB8::ABCD:0:0:100
OR	
Compressed	2001:DB8:0:0:ABCD::100

Only one :: may be used.

IPv6 Addressing Types

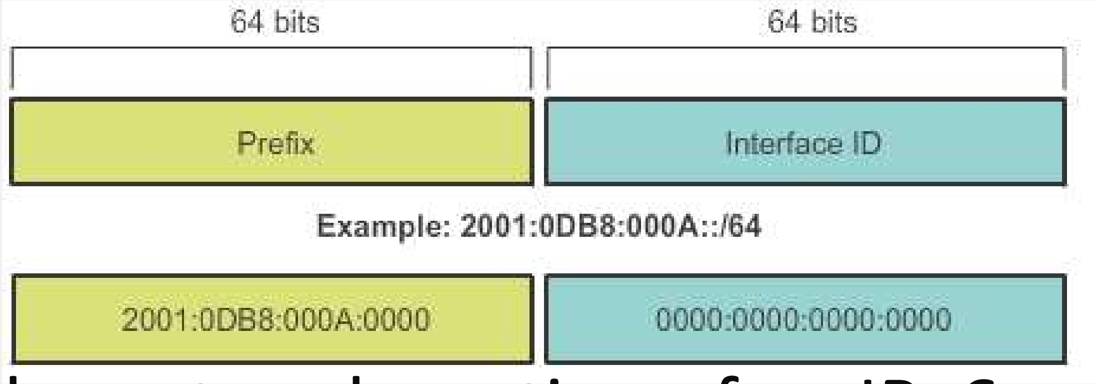
- Three types of IPv6 addressing:

- Unicast – one to one
- Multicast – one to group
- **Anycast** – first to answer

- Note: IPv6 does not have broadcast addresses.

IPv6 Prefix Length

- IPv6 does not use the dotted-decimal subnet mask notation



- Prefix length indicates the network portion of an IPv6 address using the following format:
  - IPv6 address/prefix length
  - Prefix length can range from 0 to 128
  - Typical prefix length is /64

ICMPv4 and ICMPv6 Messages

- Internet Control Messaging Protocol (ICMP) sends messages in the event of certain errors
- ICMP messages common to both ICMPv4 and ICMPv6 include:
  - Host confirmation
  - Destination or Service Unreachable
  - Time exceeded
  - Route redirection
- Commonly used for troubleshooting
  - Ping
  - Traceroute

Ping and Traceroute

- Ping sends an ICMP “echo” packet to a destination IP address, if the target address is “alive”, it should reply with an ICMP “echo-reply”

- Traceroute is similar apart from an error message is sent by each router between the sender and the target allowing the route to be identified.

Summary

- IP is connectionless and unreliable
- The network layer allows end devices to exchange data across the network.
- The network layer provides IP addressing, encapsulation and de-encapsulation
- The subnet mask (slash number or prefix length) is used to identify the network address
- IPv6 has more address space and may replace IPv4
- Ping and traceroute are useful tools

Email: k.katsifis@medcollege.edu.gr

Office Hours:

Monday 13:00-14:00

Friday 17:00-18:00